

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1516

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I_Y SRAVAN KUMAR (192472289)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____SRAVAN_____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 1: Library Book Reservation System (SIMATS Library)

Cloud Service Provider: Zoho Creator (SaaS)

Aim

To create a cloud-based Library Book Reservation System for SIMATS Library.

Fields

Field	Type
Student Name	Single Line
Register Number	Single Line
Title of Book	Single Line
Author	Single Line

Publication	Single Line
Year	Number
Edition	Single Line
Number of Copies	Number
Rack Number	Single Line
Status (Taken/Returned)	Dropdo wn
Reservation Date	Date

Workflow

1. Student enters book details.
2. System stores reservation.
3. Librarian updates status.
4. Data is saved in the cloud.

Result

A cloud-based Library Reservation System was successfully created.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 2: Payroll Processing System

Cloud Service Provider: Zoho Creator

Aim

To develop a Payroll Processing System.

Fields

Field	Type
Employee Name	Single Line
Employee ID	Single Line
Present Organization Experience	Number
Overall Experience	Number
Basic Pay	Currency

HRA	Currency
-----	----------

DA	Currency
----	----------

TA	Currency
----	----------

Gross Pay	Currency
-----------	----------

Net Pay	Currency
---------	----------

Formula (Deluge)

$\text{input.HRA} = \text{input.Basic_Pay} * 20 / 100;$

$\text{input.DA} = \text{input.Basic_Pay} * 10 / 100;$

$\text{input.TA} = \text{input.Basic_Pay} * 5 / 100;$

$\text{input.Gross_Pay} = \text{input.Basic_Pay} + \text{input.HRA} + \text{input.DA} + \text{input.TA};$

$\text{input.Net_Pay} = \text{input.Gross_Pay};$

Workflow

- Employee details entered.
- Salary calculated automatically.
- Payroll stored in cloud.

Result

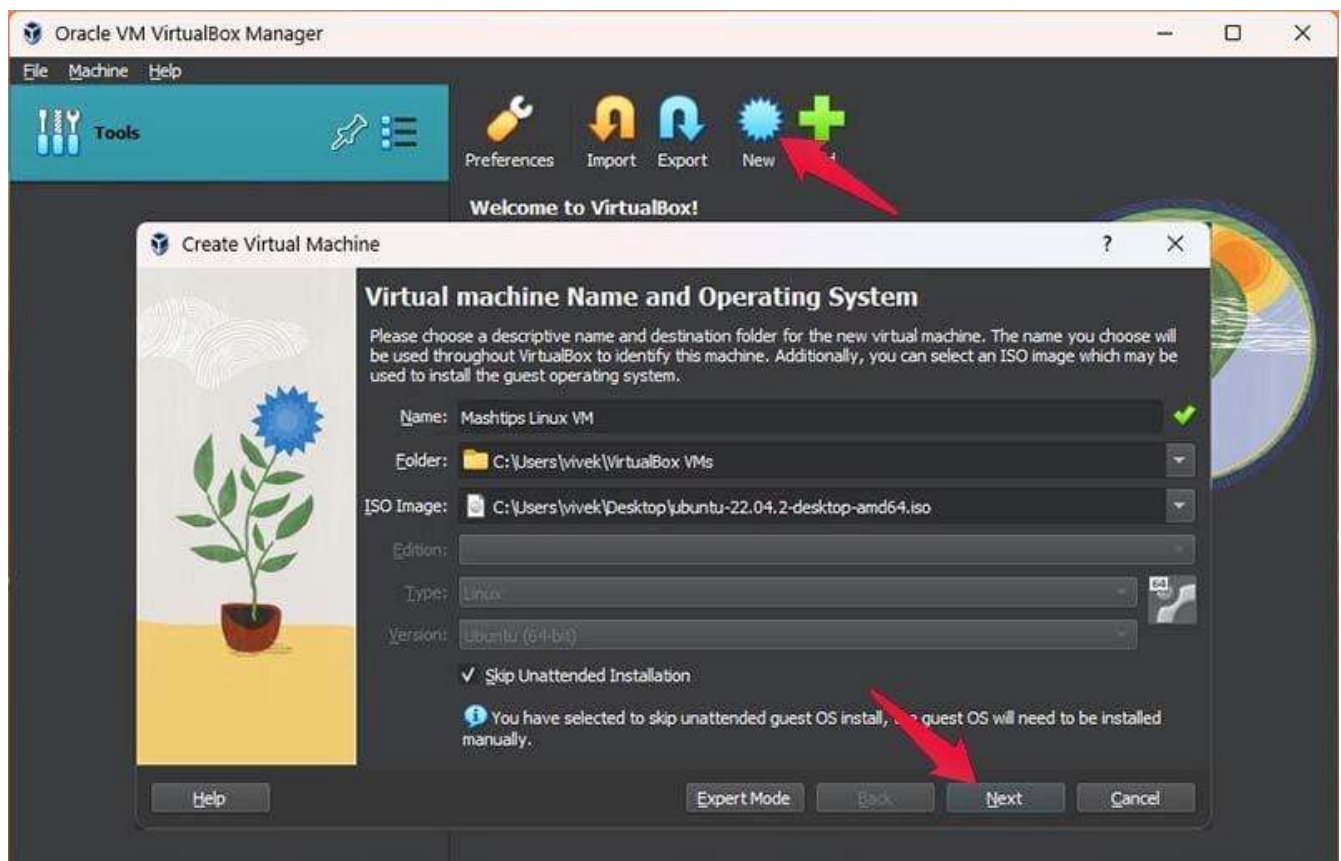
Payroll Processing System was successfully implemented.

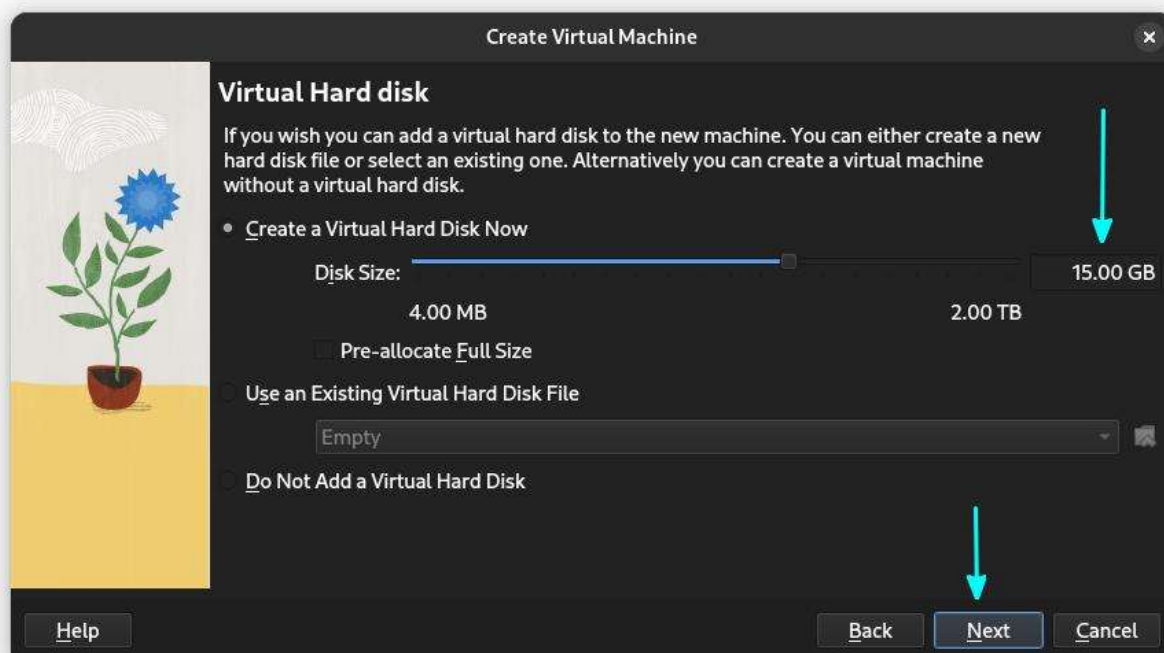
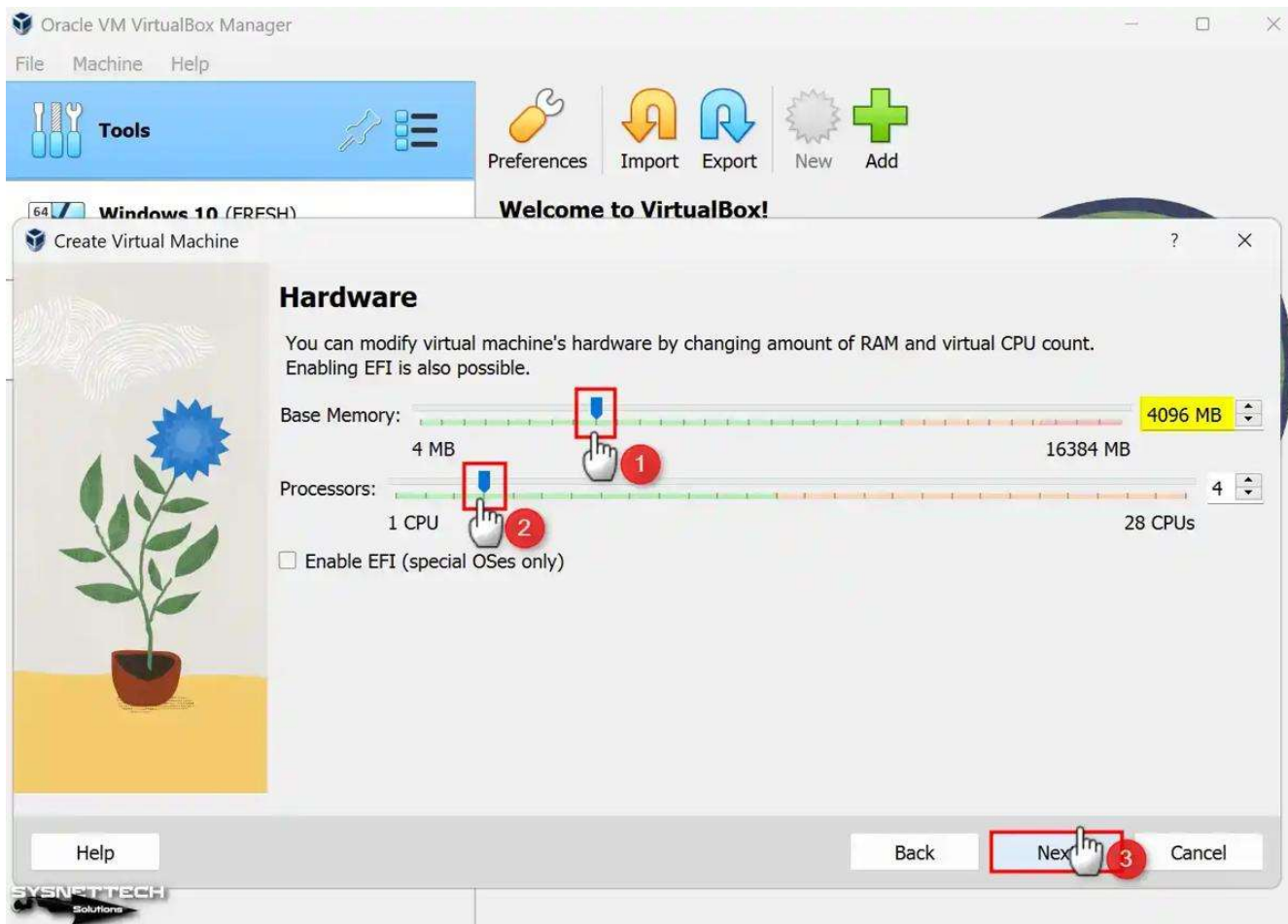
Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 3: Create Virtual Machine (1 CPU, 2GB RAM, 15GB Disk)

Aim

To install a Type-2 Hypervisor and create a Virtual Machine.





Software

- Oracle VirtualBox
- Ubuntu/Windows ISO

Configuration

Setting	Value
---------	-------

CPU	1
-----	---

RAM	2 GB
-----	------

Storage	15 GB
---------	-------

OS	Windows/Linux
----	---------------

Procedure

1. Install VirtualBox.
2. Open VirtualBox.
3. Click New.

4. Select OS.
5. Set 2 GB RAM.
6. Allocate 1 CPU.
7. Create 15 GB virtual disk.
8. Attach ISO.
9. Install OS.

Result

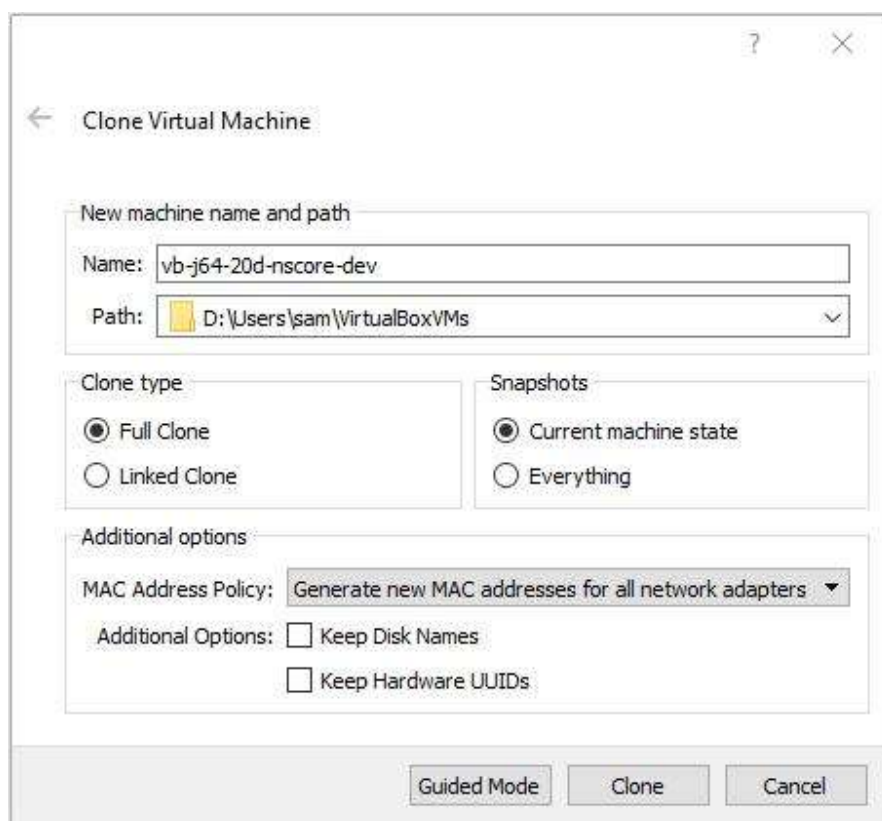
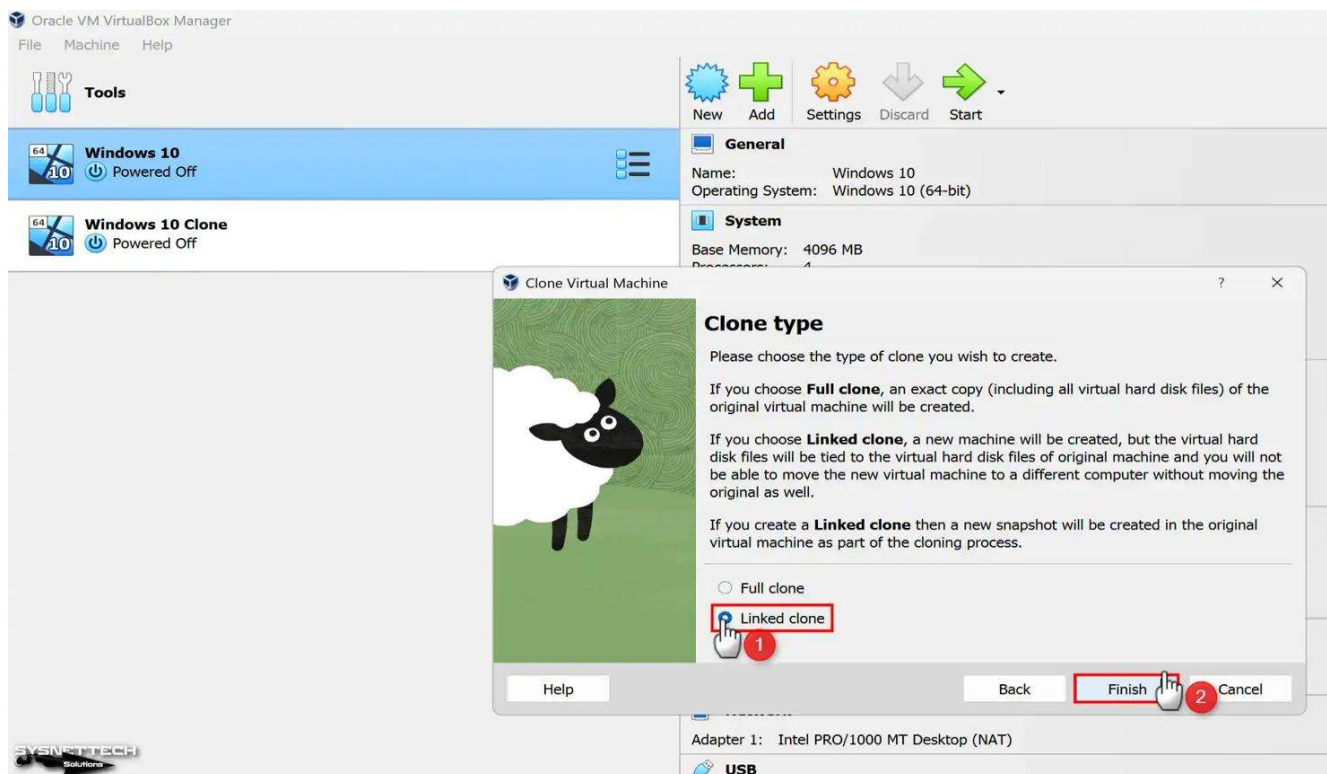
Virtual Machine was successfully created.

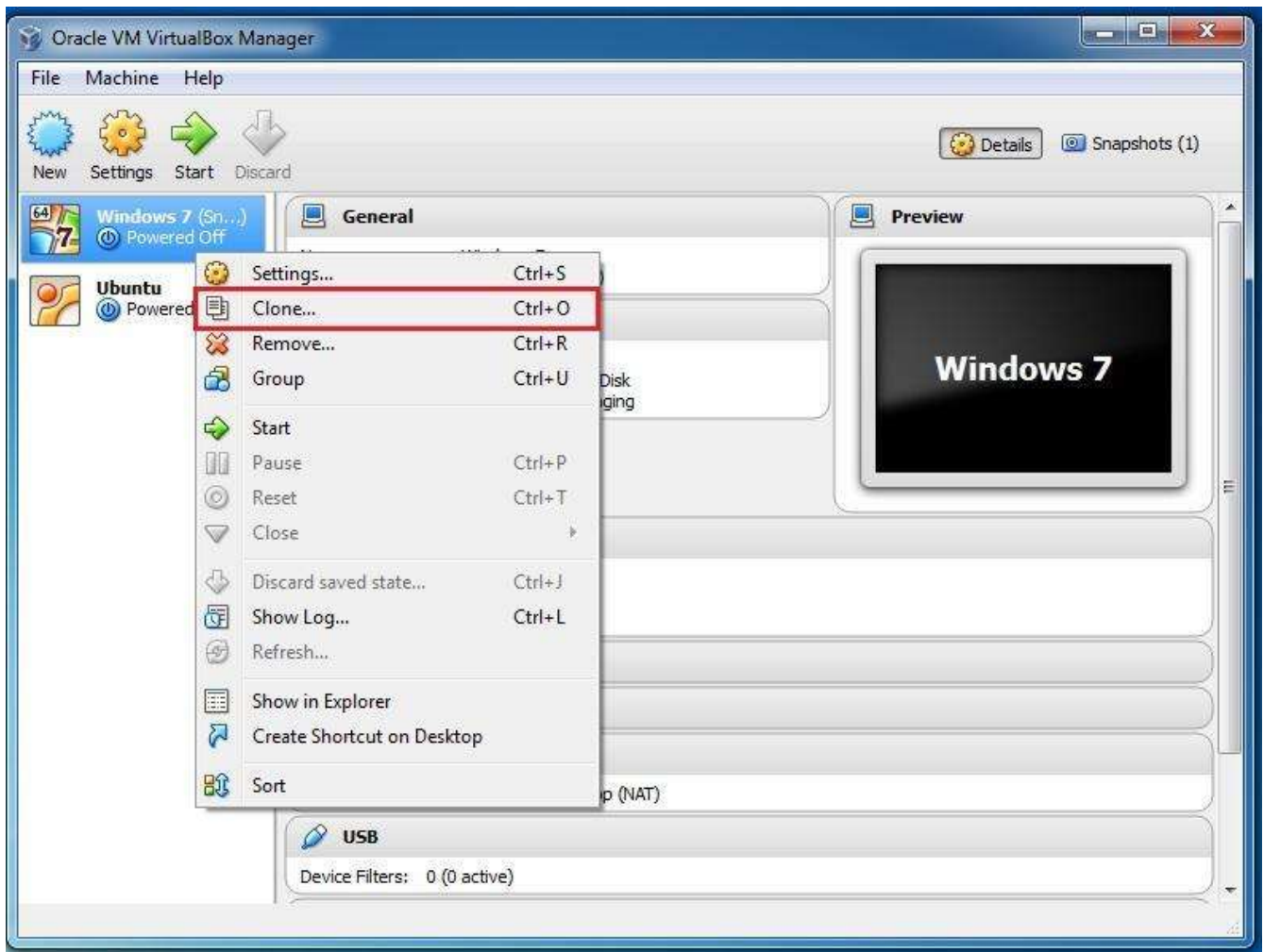
Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 4: VM Cloning

Aim

To clone an existing Virtual Machine.





5

Procedure

1. Open VirtualBox.
2. Select VM.
3. Right-click → Clone.
4. Choose Full Clone.
5. Name cloned VM.
6. Finish cloning.
7. Boot cloned VM.

Result

The cloned VM loaded successfully.

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 5: Hospital Information System (SIMATS Hospital)

Cloud Provider: Zoho Creator

Aim

To develop a Hospital Information System.

Fields

Field	Type
Patient Name	Single Line
Age	Number
Address	Multi Line

Phone	Phone
-------	-------

Email	Email
-------	-------

Attender Name	Single Line
------------------	----------------

Attender Phone	Phone
-------------------	-------

Doctor Name	Dropdo wn
----------------	--------------

Disease	Multi Line
---------	---------------

Admission Date	Date
-------------------	------

Workflow

- Patient registration.
- Doctor assignment.
- Information stored in cloud.

Result

Hospital Information System created successfully.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 6: Online Supermarket

Cloud Provider: Zoho Creator

Aim

To create an Online Supermarket application.

Customer Form

Field	Type
Customer Name	Single Line
Address	Multi Line
Phone	Phone
Email	Email
Item	Lookup

Quantity	Number
----------	--------

Total	Currency
-------	----------

Item Classification Form

Field	Type
-------	------

Item Name	Single Line
-----------	-------------

Category	Dropdown
----------	----------

Price	Currency
-------	----------

Stock	Number
-------	--------

Formula

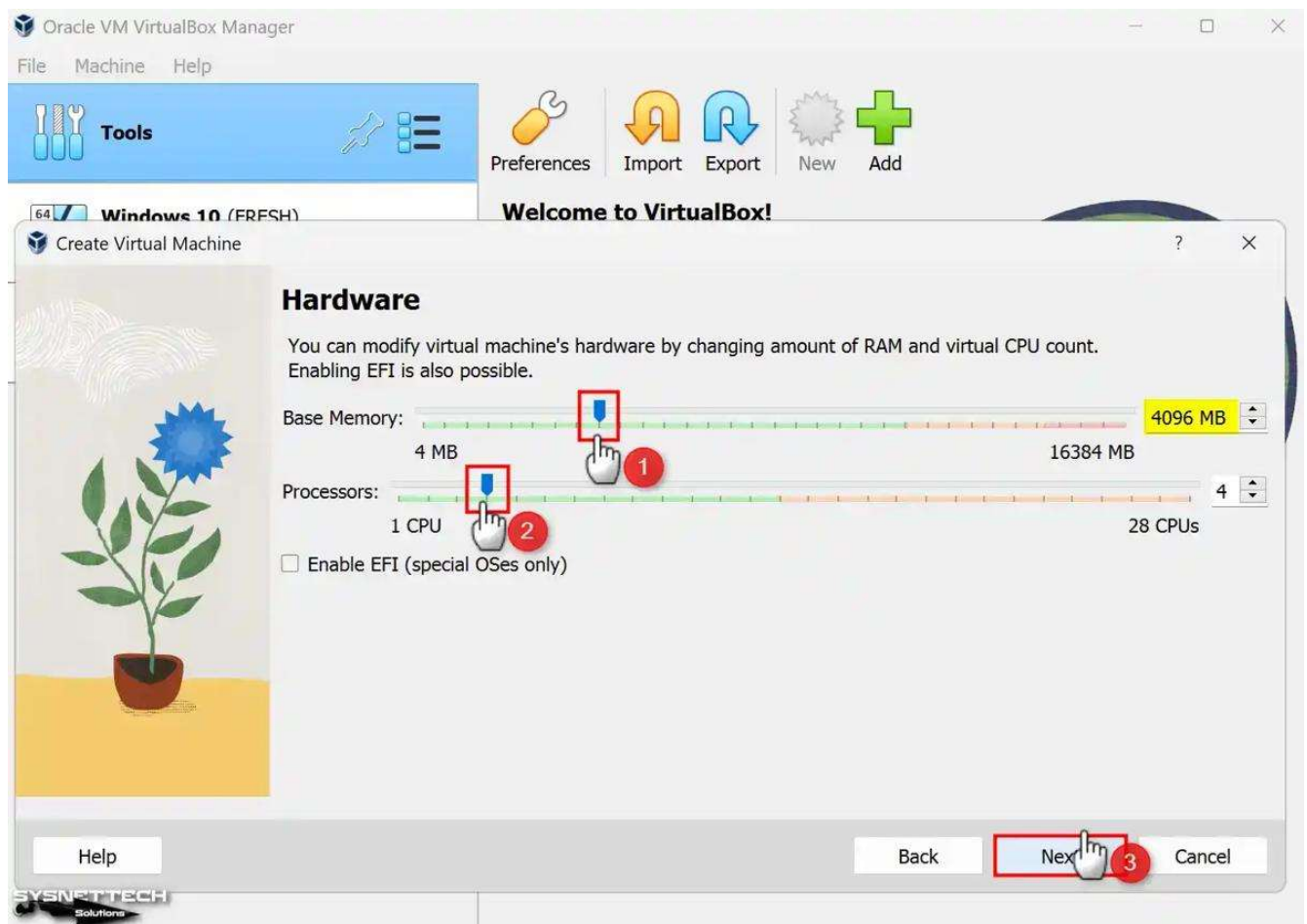
$\text{input.Total} = \text{input.Quantity} * \text{input.Item.Price};$

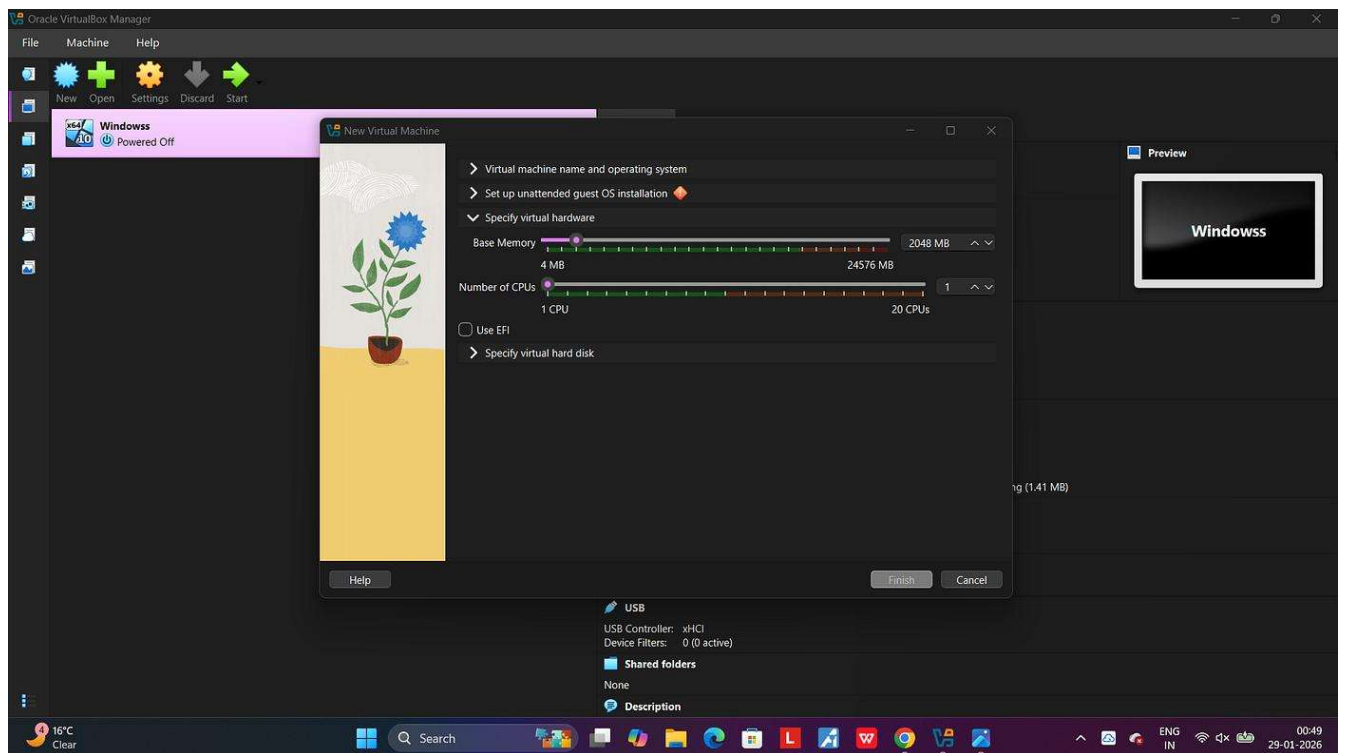
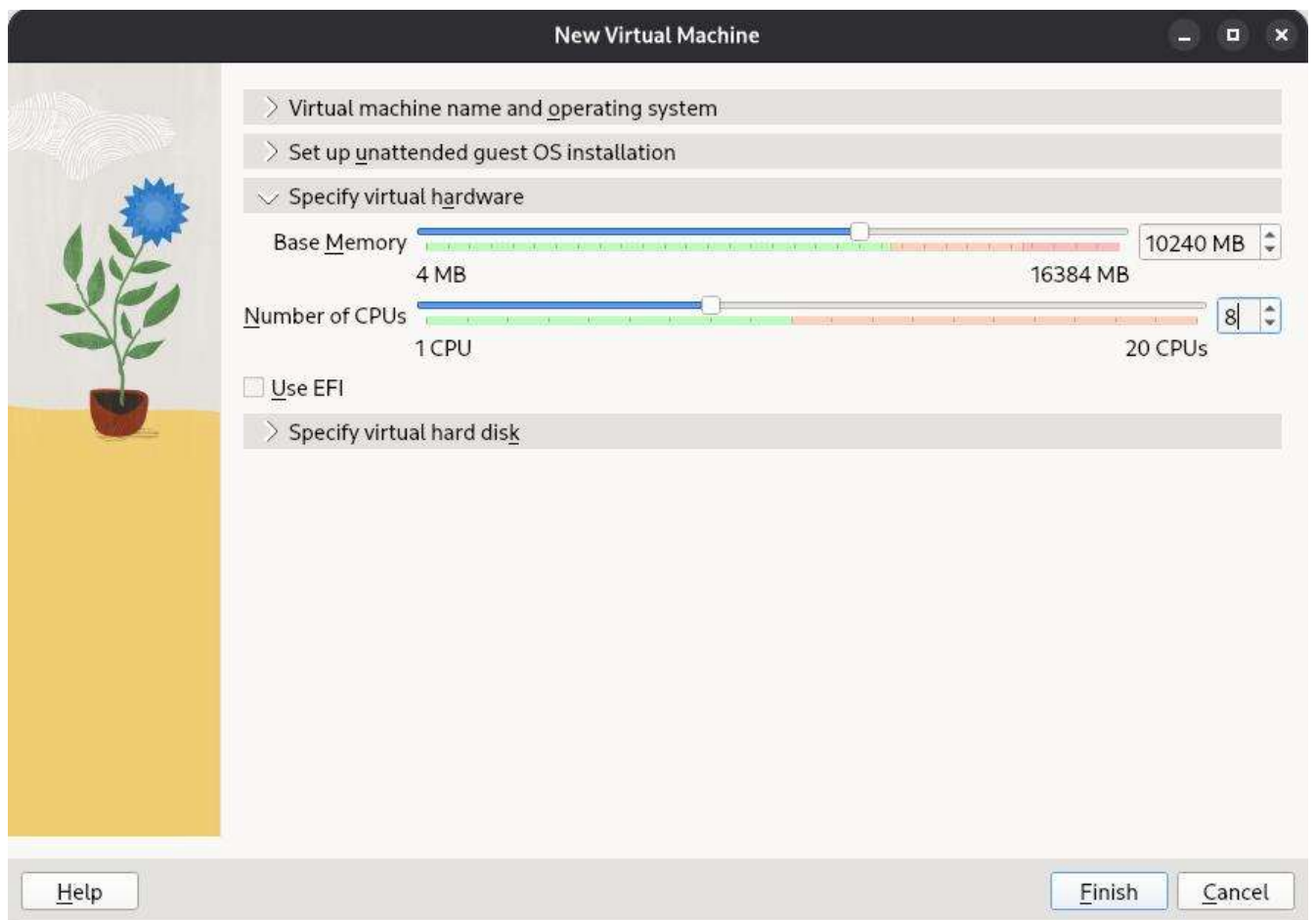
Result

Online Supermarket SaaS application created successfully.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 7: Virtual Machine (3 CPU, 10GB RAM, 10GB Disk)





Configuration

Setting Value

CPU 3

RAM 10

MB GB

Disk 10
GB

Procedure

Same as Experiment 3 with updated specifications.

Result

VM created successfully.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Experiment 9: Mark Sheet Processing System

Cloud Provider: Zoho Creator

Aim

To develop a Mark Sheet Processing System.

Fields

Field	Type
Student	Single
Name	Line
Register	Single
Number	Line
Subject 1	Numbe r
Subject 2	Numbe r

Subject 3	Number
-----------	--------

Subject 4	Number
-----------	--------

Subject 5	Number
-----------	--------

Average	Decimal
---------	---------

Percentage	Decimal
------------	---------

Rank	Number
------	--------

Performance	Dropdown
-------------	----------

Formula

```
total = input.Subject1 + input.Subject2 + input.Subject3 +  
input.Subject4 + input.Subject5;  
input.Average = total / 5;  
input.Percentage = total / 5;
```

```
if(input.Percentage >= 90)  
{  
input.Performance = "Outstanding";  
}  
else if(input.Percentage >= 75)  
{  
input.Performance = "Excellent";  
}  
else if(input.Percentage >= 50)  
{  
input.Performance = "Pass";  
}  
else  
{  
input.Performance = "Fail";  
}
```

Result

Mark Sheet Processing System created successfully

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 10: Property Buying & Rental Process (Chennai)

Cloud Provider: Zoho Creator

Aim

To create a Property Buying & Rental SaaS application.

Fields

Field	Type
Customer	Single
Name	Line

Phone	Phone
-------	-------

Email	Email
-------	-------

Property Type	Dropdo wn
------------------	--------------

Buy/Rent	Dropdo wn
----------	--------------

Commercial	Yes/No
------------	--------

Rental Agreement	Yes/No
---------------------	--------

Area	Number
------	--------

Place	Dropdo wn
-------	--------------

Range	Currenc
	y

Property	Currenc
Value	y

Formula

```
if(input.Buy_Rent == "Buy")
{
input.Property_Value = input.Range;
}
else if(input.Buy_Rent == "Rent")
{
input.Property_Value = input.Range * 12;
}
```

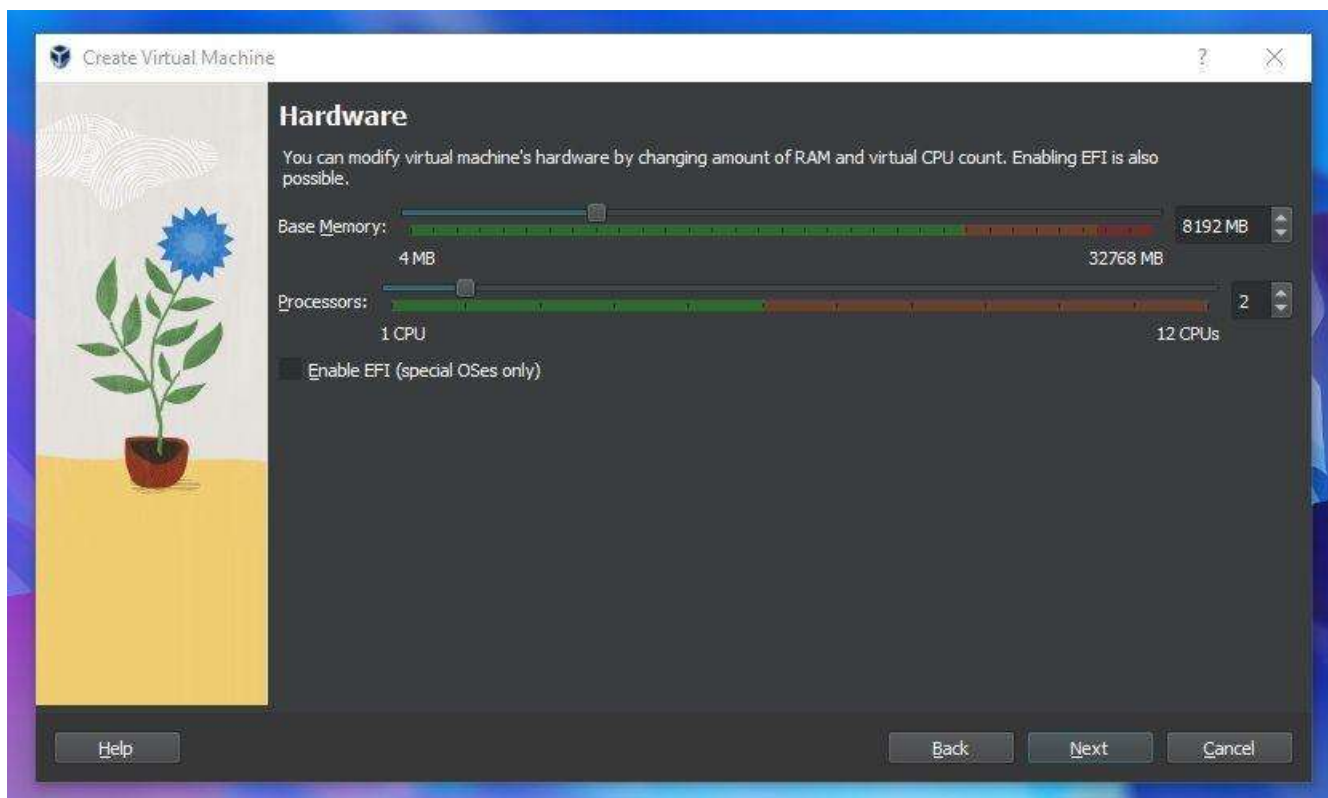
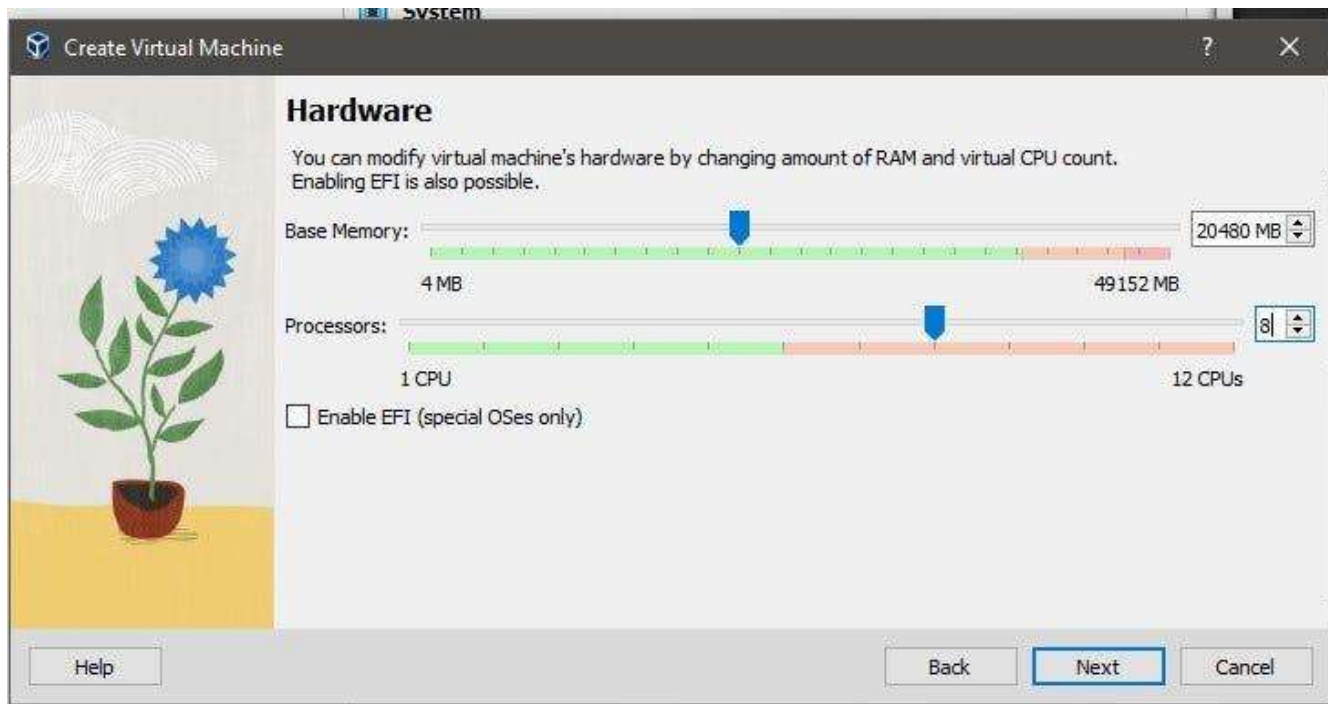
Result

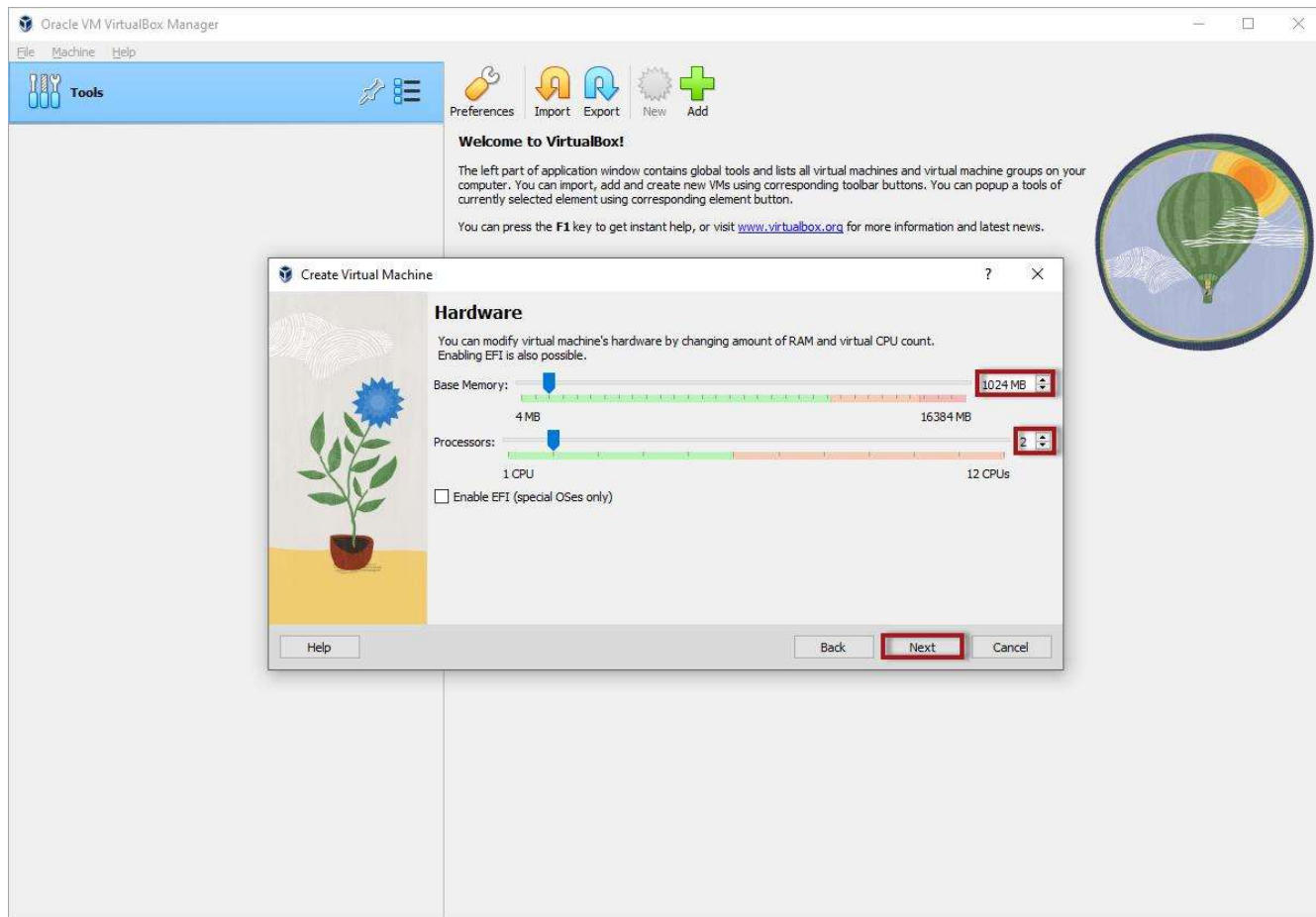
Property Buying & Rental System created successfully.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields

such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 11: Virtual Machine (5 CPU, 12GB RAM, 8GB Disk)





Configuration

Setting Value

CPU 5

RAM 12 GB

Disk 8
GB

Procedure

Follow VirtualBox VM creation with updated specifications.

Result

VM created successfully.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

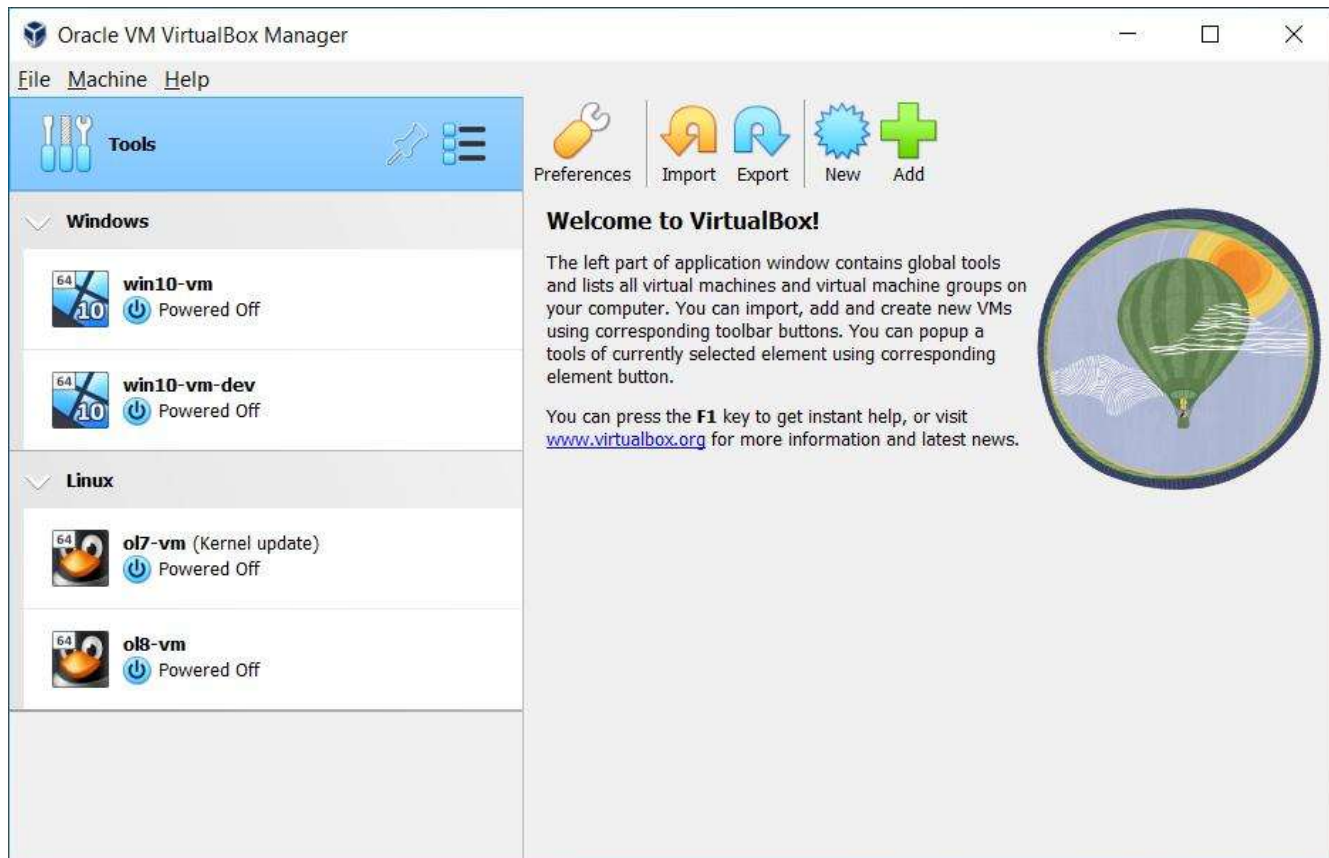
Aim

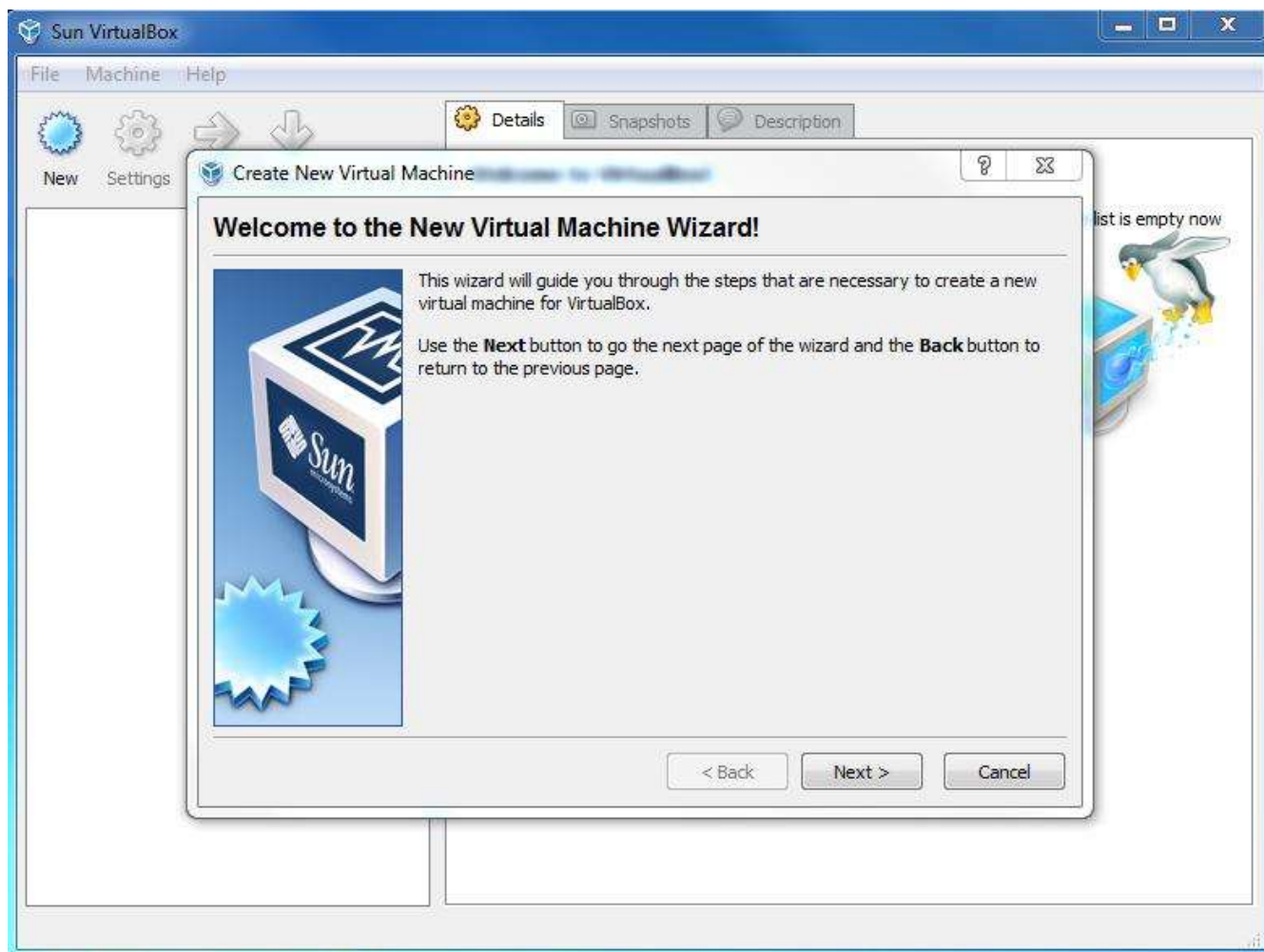
To demonstrate virtualization by installing a Type-2 Hypervisor, creating and configuring a Virtual Machine with 5 CPUs, 12 GB RAM, and 8 GB storage using Oracle VirtualBox.

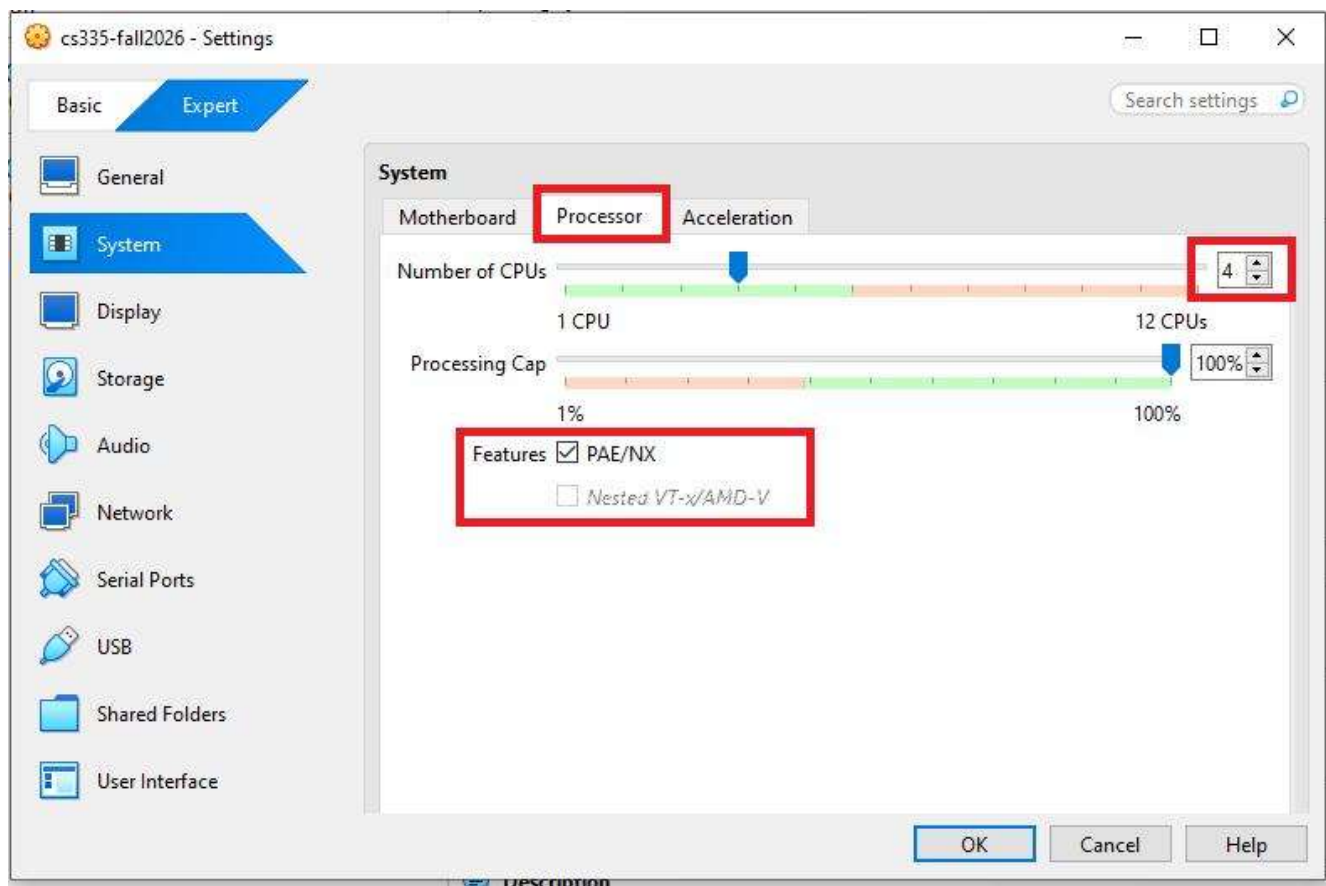
Requirements

- Host Operating System: Windows/Linux
- Type-2 Hypervisor: Oracle VirtualBox
- Guest OS ISO: Ubuntu/Linux or Windows

- Minimum host hardware supporting virtualization







5

Virtual Machine Configuration

Setting	Value
Hypervisor	Oracle VM VirtualBox
Guest OS	Ubuntu Linux (or Windows)

CPU_s 5

RAM 12 GB

Virtual 8 GB

Disk

Disk VDI (Dynamically
Type Allocated)

Procedure

1. Download and install Oracle VirtualBox.
2. Open VirtualBox and click New.
3. Enter the VM name (Example: **Ubuntu_VM**).
4. Select the operating system type and version.
5. Set Memory Size to 12288 MB (12 GB).
6. Open Settings → System → Processor.
7. Set the Processor Count to 5 CPUs.
8. Create a virtual hard disk.
9. Select VDI and Dynamically Allocated.
10. Set the disk size to 8 GB.

11. Attach the operating system ISO file through Settings → Storage.
12. Click Start and complete the OS installation.

Output

- Virtual Machine created successfully.
- Guest operating system installed and booted normally.

Result

A Virtual Machine was successfully created using Oracle VirtualBox with 5 CPUs, 12 GB RAM, and an 8 GB virtual disk.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

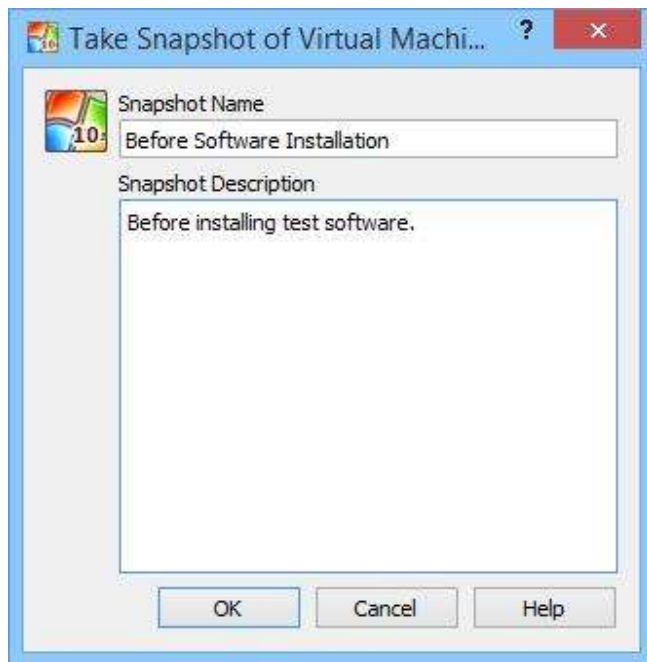
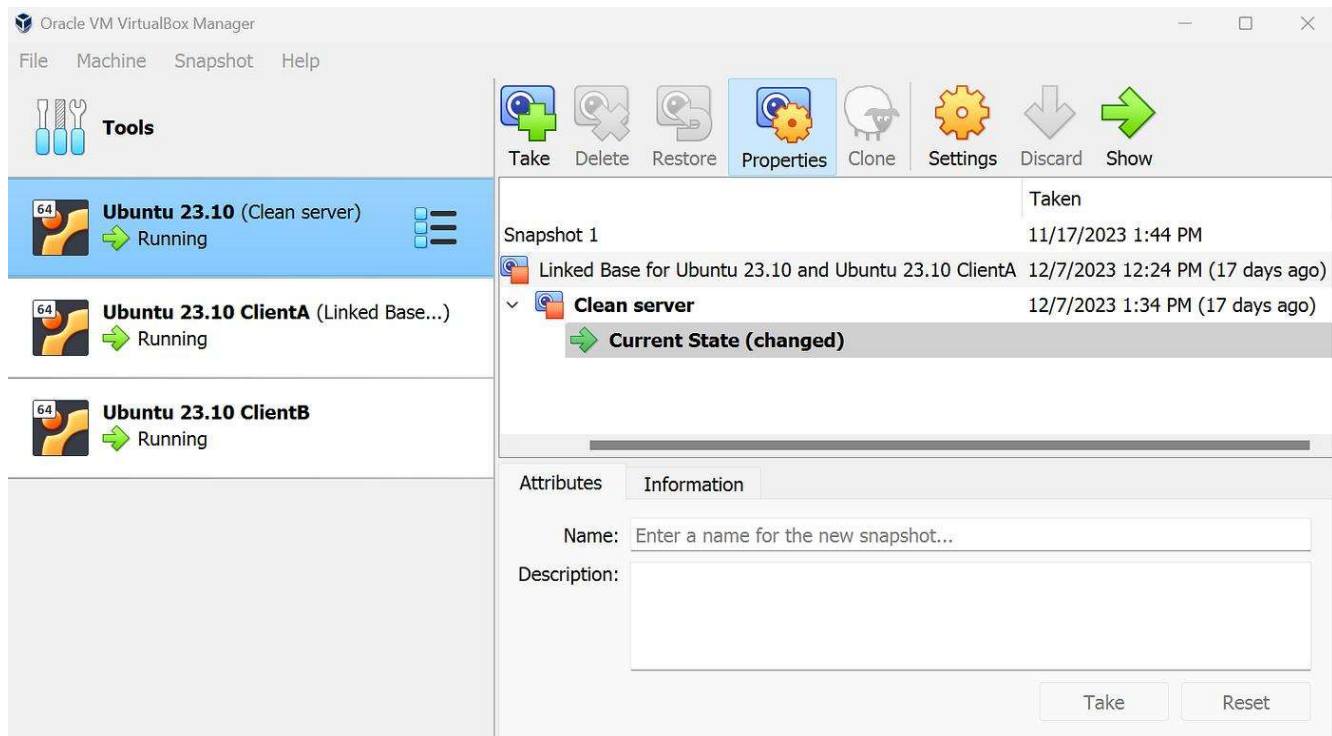
Aim

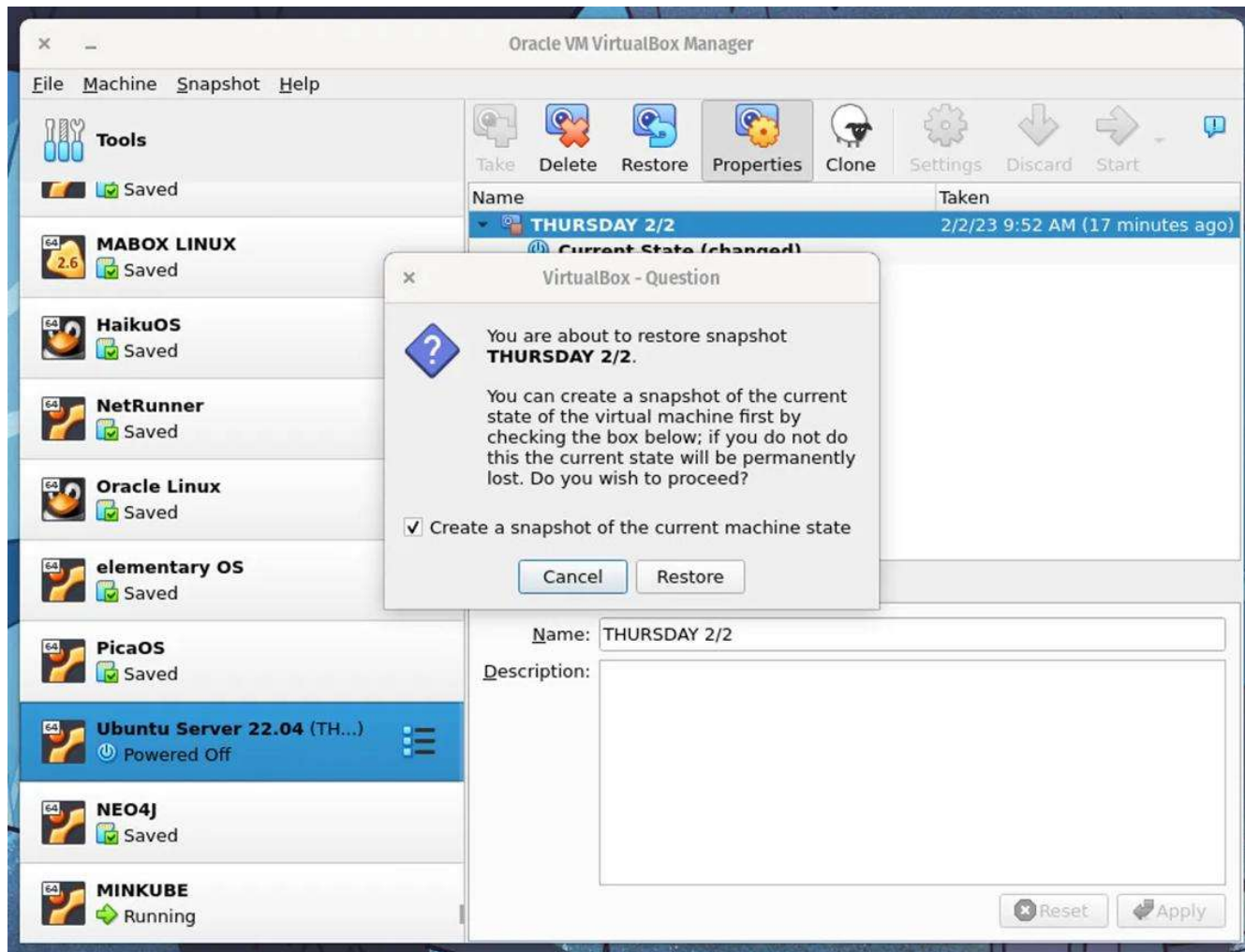
To create a snapshot of a Virtual Machine and restore it to the previous version using Oracle VirtualBox.

Requirements

- Oracle VirtualBox

- Existing Virtual Machine with Windows/Linux installed





5

Procedure

1. Open Oracle VirtualBox.
2. Select the existing Virtual Machine.
3. Start the VM and verify that it is working.
4. Shut down the VM (recommended).
5. Select the VM and open the Snapshots tab.
6. Click Take Snapshot.
7. Enter a snapshot name (Example: **Before_Update**).
8. Add an optional description.

9. Click OK.
10. Start the VM and make changes (install software or create files).
11. Shut down the VM.
12. Open the Snapshots tab again.
13. Select the **Before_Update** snapshot.
14. Click Restore Snapshot.
15. Start the VM and verify that it has returned to its previous state.

Snapshot Workflow

Verification

After restoring:

- Installed software added after the snapshot is removed.
- Files created after the snapshot disappear.
- System settings return to the snapshot state.

Difference Between Clone and Snapshot

Snapshot

Clone

Saves VM state	Creates	a
	complete copy	

Uses less storage	Uses more storage
-------------------	-------------------

Quick recovery	Independent VM
----------------	----------------

Depends on original VM	Works separately
------------------------	------------------

Result

A snapshot of the Virtual Machine was successfully created and restored, demonstrating the ability to return the system to a previous working state.

Experiment 1: Library Book Reservation System

Cloud Service Provider: Zoho Creator (SaaS)

Aim

To create a cloud-based **Library Book Reservation System** for the SIMATS Library using **Zoho Creator** and demonstrate the Software as a Service (SaaS) model.

Fields

Field Name	Field Type
Student Name	Single Line
Register Number	Single Line
Title of Book	Single Line
Author	Single Line
Publication	Single Line
Year	Number
Edition	Single Line
Number of Copies	Number
Rack Number	Single Line

Field Name	Field Type
------------	------------

Status (Taken/Returned)	Dropdown
-------------------------	----------

Reservation Date	Date
------------------	------

Procedure

1. Open **Zoho Creator** and sign in to your account.
2. Create a new application for the **Library Book Reservation System**.
3. Create a new form named **Library Book Reservation**.
4. Add the required fields:
 - Student Name
 - Register Number
 - Title of Book
 - Author
 - Publication
 - Year
 - Edition
 - Number of Copies
 - Rack Number
 - Status
 - Reservation Date
5. Set the **Status** field as a dropdown with the options:

- Taken
 - Returned
6. Set the **Reservation Date** field as a date field.
 7. Save and publish the form.
 8. Enter sample student and book details into the form.
 9. The submitted reservation details are stored in the cloud.
 10. The librarian can view the records and update the book status when the book is taken or returned.

Sample Input

Field	Sample Value
Student Name	Sravan Kumar
Register Number	192472289
Title of Book	Introduction to Artificial Intelligence
Author	Russell and Norvig
Publication	Pearson
Year	2023
Edition	4th Edition
Number of Copies	2

Field	Sample Value
-------	--------------

Rack Number	A-12
-------------	------

Status	Taken
--------	-------

Reservation Date	23-08-2026
------------------	------------

Expected Output

The system displays the submitted library reservation record:

Library Book Reservation

Student Name : Sravan Kumar

Register Number : 192472289

Title of Book : Introduction to Artificial Intelligence

Author : Russell and Norvig

Publication : Pearson

Year : 2023

Edition : 4th Edition

Number of Copies : 2

Rack Number : A-12

Status : Taken

Reservation Date : 23-08-2026

The record is stored in the **Zoho Creator cloud application**, where the librarian can view and update the reservation status.

Workflow

Student



Enter Book and Student Details



Submit Reservation



Data Stored in Cloud



Librarian Views Reservation



Update Status

(Taken / Returned)

Result

A cloud-based **Library Book Reservation System** was successfully created using **Zoho Creator as a SaaS platform**. The system stores student and book reservation details in the cloud and allows the librarian to manage the reservation status.

Experiment 2: Payroll Processing System

Cloud Service Provider: Zoho Creator (SaaS)

Aim

To develop a cloud-based **Payroll Processing System** using Zoho Creator to calculate and manage employee salary details automatically.

Fields

Field Name	Field Type
Employee Name	Single Line
Employee ID	Single Line
Present Organization Experience Number	
Overall Experience	Number
Basic Pay	Currency
HRA	Currency
DA	Currency
TA	Currency
Gross Pay	Currency
Net Pay	Currency

Procedure

1. Open **Zoho Creator** and sign in to your account.
2. Create a new application named **Payroll Processing System**.

3. Create a form named **Employee Payroll Details**.
4. Add the required employee fields.
5. Set the experience fields as **Number** fields.
6. Set Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay as **Currency** fields.
7. Create a workflow to automatically calculate HRA, DA, TA, Gross Pay, and Net Pay.
8. Enter the employee's name, ID, experience, and basic pay.
9. The system automatically calculates the salary components.
10. The calculated payroll information is stored in the cloud.

Salary Calculation

The following formulas are used:

$$\text{HRA} = \text{Basic Pay} \times 20 / 100$$

$$\text{DA} = \text{Basic Pay} \times 10 / 100$$

$$\text{TA} = \text{Basic Pay} \times 5 / 100$$

$$\text{Gross Pay} = \text{Basic Pay} + \text{HRA} + \text{DA} + \text{TA}$$

$$\text{Net Pay} = \text{Gross Pay}$$

Deluge Code

Use the following Deluge script in the workflow:

```
input.HRA = input.Basic_Pay * 20 / 100;
```

$\text{input.DA} = \text{input.Basic_Pay} * 10 / 100;$

$\text{input.TA} = \text{input.Basic_Pay} * 5 / 100;$

$\text{input.Gross_Pay} = \text{input.Basic_Pay} + \text{input.HRA} + \text{input.DA} + \text{input.TA};$

$\text{input.Net_Pay} = \text{input.Gross_Pay};$

Sample Input

Field	Sample Value
--------------	---------------------

Employee Name	Rahul Kumar
---------------	-------------

Employee ID	EMP001
-------------	--------

Present Organization Experience 2 years

Overall Experience	4 years
--------------------	---------

Basic Pay	₹30,000
-----------	---------

The remaining salary fields are calculated automatically.

Calculation

Basic Pay = ₹30,000

$\text{HRA} = 30,000 \times 20 / 100$

$\text{HRA} = ₹6,000$

$\text{DA} = 30,000 \times 10 / 100$

DA = ₹3,000

TA = $30,000 \times 5 / 100$

TA = ₹1,500

Gross Pay = $30,000 + 6,000 + 3,000 + 1,500$

Gross Pay = ₹40,500

Net Pay = ₹40,500

Expected Output

Employee Name : Rahul Kumar

Employee ID : EMP001

Present Organization Experience : 2 years

Overall Experience : 4 years

Basic Pay : ₹30,000

HRA : ₹6,000

DA : ₹3,000

TA : ₹1,500

Gross Pay : ₹40,500

Net Pay : ₹40,500

Workflow

Employee Details



Enter Basic Pay



Calculate HRA, DA and TA



Calculate Gross Pay



Calculate Net Pay



Store Payroll Details in Cloud

Result

A cloud-based **Payroll Processing System** was successfully implemented using **Zoho Creator as a SaaS platform**. The system automatically calculates HRA, DA, TA, Gross Pay, and Net Pay and stores the payroll information in the cloud.

Experiment 3: Create and Configure a Virtual Machine Using Type-2 Hypervisor

Aim

To install a **Type-2 Hypervisor** and create and configure a Virtual Machine with **1 CPU, 2 GB RAM, and 15 GB storage** using Oracle VirtualBox.

Software Required

- Oracle VirtualBox
- Windows/Linux ISO file
- Host Operating System: Windows/Linux

Virtual Machine Configuration

Setting	Value
Hypervisor	Oracle VirtualBox
Virtual CPU	1 CPU
RAM	2 GB
Storage	15 GB
Guest OS	Windows/Linux

Procedure

1. Download and install **Oracle VirtualBox** on the host computer.
2. Open **Oracle VirtualBox**.
3. Click the **New** button to create a new virtual machine.
4. Enter a suitable name for the virtual machine.
5. Select the required operating system and version.
6. Allocate **2 GB RAM** to the virtual machine.
7. Set the processor configuration to **1 CPU**.
8. Create a new virtual hard disk.
9. Allocate **15 GB** of storage to the virtual hard disk.
10. Attach the required Windows/Linux ISO file to the virtual machine.
11. Start the virtual machine.
12. Follow the operating system installation instructions.

13. After installation, restart the virtual machine and verify that the guest operating system boots successfully.

14. The virtual machine can now be used independently from the host operating system.

Configuration

Virtual Machine Name : Cloud_VM

CPU : 1 CPU

RAM : 2 GB

Storage : 15 GB

Operating System : Windows/Linux

Hypervisor : Oracle VirtualBox

Expected Output

The Oracle VirtualBox window displays the created virtual machine:

Cloud_VM

Status : Running

CPU : 1 CPU

Memory : 2048 MB

Storage : 15 GB

OS : Windows/Linux

The selected operating system starts successfully inside the virtual machine.

Result

A **Type-2 virtual machine** was successfully created and configured using **Oracle VirtualBox** with **1 CPU**, **2 GB RAM**, and **15 GB storage**.

Experiment 4: VM Cloning

Aim

To create a **clone of an existing Virtual Machine** using a Type-2 hypervisor and verify that the cloned virtual machine can be loaded successfully.

Software Required

- Oracle VirtualBox
- An existing Virtual Machine

Procedure

1. Open **Oracle VirtualBox** on the host computer.
2. Select the existing Virtual Machine that needs to be cloned.
3. Right-click on the selected Virtual Machine.
4. Select the **Clone** option.
5. Enter a suitable name for the cloned Virtual Machine.
6. Select **Full Clone** as the cloning type.
7. Complete the cloning process by clicking **Finish**.
8. Wait until the cloning process is completed.
9. The cloned Virtual Machine will appear in the VirtualBox Manager.
10. Select the cloned Virtual Machine and click **Start**.
11. Verify that the cloned Virtual Machine boots successfully.
12. Check that the operating system and previously configured settings are available in the cloned VM.

Configuration

Setting	Value
Hypervisor	Oracle VirtualBox
Clone Type	Full Clone
Source	Existing Virtual Machine
Destination	New Virtual Machine
Operating System	Same as Source VM

Expected Output

The VirtualBox Manager displays both the original and cloned virtual machines.

Original VM : Cloud_VM

Cloned VM : Cloud_VM_Clone

Clone Type : Full Clone

Status : Running

The cloned VM successfully starts and loads the operating system.

Result

The existing Virtual Machine was successfully **cloned using Oracle VirtualBox**, and the cloned VM was loaded and tested successfully.